

Let $x \neq 2m\pi$ ($m \in \mathbb{Z}$) be a real number. Observe that

$$\begin{aligned} \sum_{k=1}^n e^{ikx} &= e^{ix} \cdot \sum_{k=0}^{n-1} e^{ikx} = e^{ix} \cdot \frac{1 - e^{inx}}{1 - e^{ix}} = e^{ix} \cdot \frac{e^{\frac{1}{2}inx} - e^{-\frac{1}{2}inx}}{e^{\frac{1}{2}ix} - e^{-\frac{1}{2}ix}} \\ &= e^{\frac{ix}{2}(n+1)} \cdot \frac{\sin \frac{nx}{2}}{\sin \frac{x}{2}} \end{aligned}$$

Since $e^{ikx} = \cos kx + i \cdot \sin kx$ and

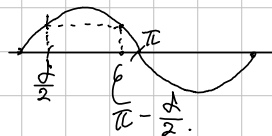
$$e^{\frac{ix}{2}(n+1)} = \cos \frac{(n+1)x}{2} + i \cdot \sin \frac{(n+1)x}{2}$$

So therefore

$$\sum_{k=1}^n \cos kx = \frac{\sin \frac{nx}{2}}{\sin \frac{x}{2}} \cdot \cos \frac{(n+1)x}{2} \quad \text{and} \quad \sum_{k=1}^n \sin kx = \frac{\sin \frac{nx}{2}}{\sin \frac{x}{2}} \cdot \sin \frac{(n+1)x}{2}$$

Now, for fixed $0 < \delta < \pi$, given $x \in [\delta, 2\pi - \delta]$,

$$\left| \sum_{k=1}^n \cos kx \right|, \left| \sum_{k=1}^n \sin kx \right| \leq \left| \frac{1}{\sin \frac{x}{2}} \right| \leq \left| \frac{1}{\sin \frac{\delta}{2}} \right|$$



Consequently, we obtain a Theorem:

[Thm. Suppose that $\{a_n\}$ converges to 0, and $\sum |a_n - a_{n-1}|$ converges.

Given $\delta \in (0, \pi)$, $\sum_{n=1}^{\infty} a_n \cdot \sin nx$ and $\sum_{n=1}^{\infty} a_n \cdot \cos nx$ converges uniformly on $[\delta, 2\pi - \delta]$.

Proof.